SHIVESH HARAN P

AI/ML & Full-Stack Developer · B.Tech CSE, SRM KTR
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PROFESSIONAL SUMMARY

Final-year Computer Science undergraduate (CGPA 8.45) who builds and ships end-to-end AI applications, from speech-recognition pipelines and neuro-symbolic diagnosis with RAG to simulation systems. Hands-on across Python, FastAPI, React/TypeScript and MongoDB, with a focus on tested, documented, deployable code. Seeking AI/ML engineering and software development roles.

EDUCATION

SRM Institute of Science and Technology, KTR Campus, Chennai **CGPA: 8.45**
Bachelor of Technology in Computer Science and Engineering
St. Marks Matriculation Higher Secondary School, Chennai Class XII: 75% · Class X: 100%

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, TypeScript, C#
AI / ML: Scikit-learn, PyTorch, Pandas, NumPy, NLP, RAG, Speech Recognition (Whisper), Classification, Regression, Clustering
Web & Backend: FastAPI, Flask, React, Vite, Tailwind CSS, REST APIs, JWT Authentication
Databases: MongoDB, MySQL
Tools & Platforms: Git, GitHub, Docker, FFmpeg, Unity 6, AWS, Power BI

PROJECTS

LectureMind AI | *AI Lecture Transcription & Knowledge Extraction* Live · GitHub **Oct 2026**
React, TypeScript, Tailwind, FastAPI, MongoDB, FFmpeg, faster-whisper, OpenAI API

- Built a full-stack platform that turns lecture audio/video into timestamped transcripts, summaries, smart notes, quizzes, flashcards and an interactive knowledge-map graph.
- Engineered a background processing pipeline: FFmpeg audio extraction, faster-whisper ASR, then chunked map-reduce summarisation with schema-validated JSON outputs, with per-step status and retries persisted in MongoDB.
- Implemented grounded lecture chat with retrieval and timestamp citations, plus a content-safety layer flagging profanity, insults and threats (English, Hinglish, Tamil-English) with severity levels.
- Secured with JWT httpOnly cookies, bcrypt and per-user ownership checks; covered by 82 automated pytest tests and deployed as a live demo.

NeuroSy-RAG | *Hybrid Neuro-Symbolic Healthcare Platform* Live · GitHub **Jan 2026 – Mar 2026**
Python, Scikit-learn, NLP, RAG, FAISS, FastAPI, Streamlit, OpenStreetMap APIs

- Engineered a neuro-symbolic diagnostic pipeline in which a Random Forest + Decision Tree ensemble proposes conditions and a knowledge graph with a deterministic rule engine validates them, abstaining instead of guessing to curb hallucinations.
- Developed NLP entity extraction with negation handling, typo tolerance and duration rules, mapping free-text patient input to structured symptoms across 34 conditions and 79 symptoms.
- Built a RAG layer over a curated medical knowledge base (TF-IDF / MiniLM + FAISS) that generates cited explanations, and ranked nearby hospitals with geospatial APIs and weighted scoring.

AI-Powered Smart City Digital Twin GitHub **Oct 2026**
Unity 6, C#, Universal Render Pipeline, NUnit

- Built a procedurally generated 3D city with a live traffic simulation and A* route planning that reroutes vehicles around incidents and congestion.
- Designed an adaptive, queue-based traffic-signal optimizer with A/B comparison against fixed timing, plus incident simulation and emergency-vehicle dispatch.
- Added pollution and energy monitoring layers with trend forecasting and a control-room dashboard for decision support; core logic verified by 28 NUnit tests.

EXPERIENCE

UI/UX Intern **May 2025 – Jul 2025**

- Designed wireframes and interactive prototypes for user-centred digital products.
- Created clean, consistent and engaging interface layouts with close attention to visual detail.
- Iterated quickly on feedback, picking up new design tools and workflows to improve the user experience.

CERTIFICATIONS

Machine Learning (NPTEL) | UI/UX | Networking Basics (Cisco) | Python 300+ Exercises (Udemy)